Checking Sensitivities for Different Off-axis Angles

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Allegati: oa_optimization.pdf (582 KB)

Dear VALOR group,

I was curious if the choice of a 2.5 degree off-axis angle for both T2K and T2HK is still optimal considering what we now know about theta_13, NC backgrounds and the fraction of antineutrino mode running. I put together a few slides (attached) giving a hand-waiving argument for why a smaller off-axis angle could be more optimal. I also produced some flux weight histograms to reweight the 2.5 degree flux to 2.0 or 2.25 degrees. The histogram file is also described in the attached slides. Would it be possible to use these flux weight histograms in the VALOR framework to check how oscillation parameter sensitivities are affected by the choice of off-axis angle? To use them, you would just apply all weights as usual and then multiply by these weights as well.

Thanks,
Mark

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